

100 Days of C Programming

A practical day-by-day coding journey • Day 1 → Day 100

Day 1 - Hello World

```
#include <stdio.h>
int main(){ printf("Hello World"); return 0; }
```

Day 2 - Print Name

```
#include <stdio.h>
int main(){ printf("My Name is Aryash"); return 0; }
```

Day 3 - Variables

```
#include <stdio.h>
int main(){ int age=20; printf("%d",age); return 0; }
```

Day 4 - Data Types

```
#include <stdio.h>
int main(){ int a=10; float b=5.5; char c='A'; printf("%d %.1f %c",a,b,c); return 0; }
```

Day 5 - User Input

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); printf("%d",n); return 0; }
```

Day 6 - Addition

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a+b); return 0; }
```

Day 7 - Subtraction

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a-b); return 0; }
```

Day 8 - Multiplication

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a*b); return 0; }
```

Day 9 - Division

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a/b); return 0; }
```

Day 10 - Modulus

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a%b); return 0; }
```

Day 11 - Even or Odd

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); printf(n%2==0?"Even":"Odd"); return 0; }
```

Day 12 - Positive or Negative

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); printf(n>=0?"Positive":"Negative"); return 0; }
```

Day 13 - Largest of Two

```
#include <stdio.h>
int main(){ int a,b; scanf("%d%d",&a,&b); printf("%d",a>b?a:b); return 0; }
```

Day 14 - Largest of Three

```
#include <stdio.h>
int main(){ int a,b,c; scanf("%d%d%d",&a,&b,&c); int m=a>b?a:b; m=m>c?m:c; printf("%d",m); return 0; }
```

Day 15 - If Statement

```
#include <stdio.h>
int main(){ int n=10; if(n>5) printf("Greater"); return 0; }
```

Day 16 - If Else

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); printf(n>=18?"Adult":"Minor"); return 0; }
```

Day 17 - Else If Ladder

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); if(n>=90)printf("A");else if(n>=60)printf("B");else printf("C"); return 0; }
```

Day 18 - Switch

```
#include <stdio.h>
int main(){ int n; scanf("%d",&n); switch(n){case 1:printf("Monday");break;case 2:printf("Tuesday");break;default:printf("Other Day");break;} return 0; }
```

Day 19 - For Loop

```
#include <stdio.h>
int main(){ for(int i=1;i<=10;i++) printf("%d ",i); return 0; }
```

Day 20 - While Loop

```
#include <stdio.h>
int main(){ int i=1; while(i<=10){printf("%d ",i);i++;} return 0; }
```

Day 21 - Do While

```
#include <stdio.h>
int main(){ int i=1; do{printf("%d ",i);i++;}while(i<=10); return 0; }
```

Day 22 - Break

```
#include <stdio.h>
int main(){ for(int i=1;i<=10;i++){if(i==5)break;printf("%d ",i);} return 0; }
```

Day 23 - Continue

```
#include <stdio.h>
int main(){ for(int i=1;i<=5;i++){if(i==3)continue;printf("%d ",i);} return 0; }
```

Day 24 - Nested Loop

```
#include <stdio.h>
int main(){ for(int i=1;i<=3;i++){for(int j=1;j<=3;j++)printf("* ");printf("\n");} return 0; }
```

Day 25 - Star Pattern

```
#include <stdio.h>
int main(){ for(int i=1;i<=5;i++){for(int j=1;j<=i;j++)printf("* ");printf("\n");} return 0; }
```

Day 26 - Number Pattern

```
#include <stdio.h>
int main(){ for(int i=1;i<=5;i++){for(int j=1;j<=i;j++)printf("%d ",j);printf("\n");} return 0; }
```

Day 27 - Reverse Pattern

```
#include <stdio.h>
int main(){ for(int i=5;i>=1;i--){for(int j=1;j<=i;j++)printf("* ");printf("\n");} return 0; }
```

Day 28 - Function

```
#include <stdio.h>
void hello(){printf("Hello");}
int main(){hello();return 0;}
```

Day 29 - Function with Parameter

```
#include <stdio.h>
void add(int a,int b){printf("%d",a+b);}
int main(){add(10,20);return 0;}
```

Day 30 - Return Function

```
#include <stdio.h>
int add(int a,int b){return a+b;}
int main(){printf("%d",add(10,20));return 0;}
```

Day 31 - Call by Value

```
#include <stdio.h>
void change(int x){x=100;}
int main(){int a=10;change(a);printf("%d",a);return 0;}
```

Day 32 - Recursion

```
#include <stdio.h>
void count(int n){if(n==0)return;printf("%d ",n);count(n-1);}
int main(){count(5);return 0;}
```

Day 33 - Array

```
#include <stdio.h>
int main(){int a[]={10,20,30,40,50};for(int i=0;i<5;i++)printf("%d ",a[i]);return 0;}
```

Day 34 - Array Sum

```
#include <stdio.h>
int main(){int a[]={1,2,3,4,5},sum=0;for(int i=0;i<5;i++)sum+=a[i];printf("%d",sum);return 0;}
```

Day 35 - Array Average

```
#include <stdio.h>
int main(){int a[]={10,20,30,40,50},sum=0;for(int i=0;i<5;i++)sum+=a[i];printf("%.2f",sum/5.0);return 0;}
```

Day 36 - Linear Search

```
#include <stdio.h>
int main(){int a[]={10,20,30,40},key=30;for(int i=0;i<4;i++)if(a[i]==key)printf("Found");return 0;}
```

Day 37 - Find Maximum

```
#include <stdio.h>
int main(){int a[]={10,50,20,40},m=a[0];for(int i=1;i<4;i++)if(a[i]>m)m=a[i];printf("%d",m);return 0;}
```

Day 38 - Find Minimum

```
#include <stdio.h>
int main(){int a[]={10,50,20,40},m=a[0];for(int i=1;i<4;i++)if(a[i]<m)m=a[i];printf("%d",m);return 0;}
```

Day 39 - Bubble Sort

```
#include <stdio.h>
int main(){int a[]={5,3,4,1,2};for(int i=0;i<5;i++)for(int j=0;j<4-i;j++)if(a[j]>a[j+1]){int t=a[j];a[j]=a[j+1];a[j+1]=t;}}
```

Day 40 - Selection Sort

```
#include <stdio.h>
int main(){int a[]={5,3,4,1,2};for(int i=0;i<5;i++){int m=i;for(int j=i+1;j<5;j++)if(a[j]<a[m])m=j;int t=a[i];a[i]=a[m];a[m]=t;}}
```

Day 41 - Insertion Sort

```
#include <stdio.h>
```

```
int main(){int a[]={5,3,4,1,2};for(int i=1;i<5;i++){int k=a[i],j=i-1;while(j>=0&& a[j]>k){a[j+1]=a[j];j--;}a[j+1]=k;}for
```

Day 42 - 2D Array

```
#include <stdio.h>
int main(){int a[2][2]={{1,2},{3,4}};for(int i=0;i<2;i++){for(int j=0;j<2;j++)printf("%d ",a[i][j]);printf("\n");}return
```

Day 43 - Matrix Addition

```
#include <stdio.h>
int main(){int a[2][2]={{1,2},{3,4}},b[2][2]={{5,6},{7,8}};for(int i=0;i<2;i++)for(int j=0;j<2;j++)printf("%d ",a[i][j]+b[i][j]);printf("\n");return
```

Day 44 - Matrix Transpose

```
#include <stdio.h>
int main(){int a[2][2]={{1,2},{3,4}};for(int i=0;i<2;i++){for(int j=0;j<2;j++)printf("%d ",a[j][i]);printf("\n");}return
```

Day 45 - String

```
#include <stdio.h>
int main(){char name[]="Aryash";printf("%s",name);return 0;}
```

Day 46 - String Input

```
#include <stdio.h>
int main(){char name[50];scanf("%49s",name);printf("%s",name);return 0;}
```

Day 47 - String Length

```
#include <stdio.h>
#include <string.h>
int main(){char s[]="Hello";printf("%zu",strlen(s));return 0;}
```

Day 48 - String Copy

```
#include <stdio.h>
#include <string.h>
int main(){char a[20],b[]="Hello";strcpy(a,b);printf("%s",a);return 0;}
```

Day 49 - String Compare

```
#include <stdio.h>
#include <string.h>
int main(){printf("%d",strcmp("Hello","Hello"));return 0;}
```

Day 50 - String Concatenation

```
#include <stdio.h>
#include <string.h>
int main(){char a[30]="Hello ";strcat(a,"World");printf("%s",a);return 0;}
```

Day 51 - Pointer

```
#include <stdio.h>
int main(){int a=10;int *p=&a;printf("%d",*p);return 0;}
```

Day 52 - Pointer Address

```
#include <stdio.h>
int main(){int a=10;int *p=&a;printf("%p", (void*)p);return 0;}
```

Day 53 - Pointer and Array

```
#include <stdio.h>
int main(){int a[]={10,20,30};int *p=a;printf("%d",*(p+1));return 0;}
```

Day 54 - Pointer Function

```
#include <stdio.h>
void change(int *p){*p=100;}
int main(){int a=10;change(&a);printf("%d",a);return 0;}
```

Day 55 - Swap Using Pointer

```
#include <stdio.h>
void swap(int *a,int *b){int t=*a;*a=*b;*b=t;}
int main(){int a=10,b=20;swap(&a,&b);printf("%d %d",a,b);return 0;}
```

Day 56 - Pointer Arithmetic

```
#include <stdio.h>
int main(){int a[]={10,20,30};int *p=a;printf("%d",*(p+2));return 0;}
```

Day 57 - Structure

```
#include <stdio.h>
struct Student{int age;};
int main(){struct Student s={20};printf("%d",s.age);return 0;}
```

Day 58 - Structure String

```
#include <stdio.h>
struct Student{char name[20];int age;};
int main(){struct Student s={"Aryash",20};printf("%s %d",s.name,s.age);return 0;}
```

Day 59 - Array of Structures

```
#include <stdio.h>
struct Student{int age;};
int main(){struct Student s[2]={20},{21};printf("%d %d",s[0].age,s[1].age);return 0;}
```

Day 60 - Nested Structure

```
#include <stdio.h>
struct Address{int pin;};
struct Student{struct Address a;};
int main(){struct Student s={{380001}};printf("%d",s.a.pin);return 0;}
```

Day 61 - Union

```
#include <stdio.h>
union Data{int x;float y;};
int main(){union Data d;d.x=10;printf("%d",d.x);return 0;}
```

Day 62 - Enum

```
#include <stdio.h>
enum Day{MON,TUE,WED};
int main(){enum Day d=MON;printf("%d",d);return 0;}
```

Day 63 - Typedef

```
#include <stdio.h>
typedef int Number;
int main(){Number n=10;printf("%d",n);return 0;}
```

Day 64 - File Writing

```
#include <stdio.h>
int main(){FILE *f=fopen("data.txt","w");fprintf(f,"Hello C");fclose(f);return 0;}
```

Day 65 - File Reading

```
#include <stdio.h>
int main(){FILE *f=fopen("data.txt","r");int c;while((c=fgetc(f))!=EOF)putchar(c);fclose(f);return 0;}
```

Day 66 - File Append

```
#include <stdio.h>
int main(){FILE *f=fopen("data.txt","a");fprintf(f,"\nNew Data");fclose(f);return 0;}
```

Day 67 - malloc()

```
#include <stdio.h>
#include <stdlib.h>
int main(){int *p=malloc(5*sizeof(int));p[0]=10;printf("%d",p[0]);free(p);return 0;}
```

Day 68 - calloc()

```
#include <stdio.h>
#include <stdlib.h>
int main(){int *p=calloc(5, sizeof(int));printf("%d",p[0]);free(p);return 0;}
```

Day 69 - realloc()

```
#include <stdio.h>
#include <stdlib.h>
int main(){int *p=malloc(2*sizeof(int));p=realloc(p,5*sizeof(int));free(p);return 0;}
```

Day 70 - free()

```
#include <stdlib.h>
int main(){int *p=malloc(sizeof(int));*p=10;free(p);return 0;}
```

Day 71 - Command Line Argument

```
#include <stdio.h>
int main(int argc,char *argv[]){printf("%d",argc);return 0;}
```

Day 72 - Factorial

```
#include <stdio.h>
int main(){int n=5,f=1;for(int i=1;i<=n;i++)f*=i;printf("%d",f);return 0;}
```

Day 73 - Fibonacci

```
#include <stdio.h>
int main(){int a=0,b=1;for(int i=0;i<10;i++){printf("%d ",a);int c=a+b;a=b;b=c;}return 0;}
```

Day 74 - Prime Number

```
#include <stdio.h>
int main(){int n=17,p=1;for(int i=2;i<n;i++)if(n%i==0)p=0;printf(p?"Prime":"Not Prime");return 0;}
```

Day 75 - Reverse Number

```
#include <stdio.h>
int main(){int n=123,r=0;while(n){r=r*10+n%10;n/=10;}printf("%d",r);return 0;}
```

Day 76 - Palindrome Number

```
#include <stdio.h>
int main(){int n=121,t=n,r=0;while(n){r=r*10+n%10;n/=10;}printf(t==r?"Palindrome":"Not Palindrome");return 0;}
```

Day 77 - Armstrong Number

```
#include <stdio.h>
int main(){int n=153,t=n,s=0;while(n){int d=n%10;s+=d*d*d;n/=10;}printf(s==t?"Armstrong":"Not Armstrong");return 0;}
```

Day 78 - GCD

```
#include <stdio.h>
int main(){int a=12,b=18;while(b){int t=b;b=a%b;a=t;}printf("%d",a);return 0;}
```

Day 79 - LCM

```
#include <stdio.h>
int main(){int a=12,b=18,x=a,y=b;while(y){int t=y;y=x%y;x=t;}printf("%d",(a*b)/x);return 0;}
```

Day 80 - Sum of Digits

```
#include <stdio.h>
int main(){int n=123,s=0;while(n){s+=n%10;n/=10;}printf("%d",s);return 0;}
```

Day 81 - Count Digits

```
#include <stdio.h>
int main(){int n=12345,c=0;while(n){c++;n/=10;}printf("%d",c);return 0;}
```

Day 82 - Stack

```
#include <stdio.h>
int main(){int stack[5],top=-1;stack[++top]=10;stack[++top]=20;printf("%d",stack[top--]);return 0;}
```

Day 83 - Queue

```
#include <stdio.h>
int main(){int q[5],front=0,rear=0;q[rear++]=10;q[rear++]=20;printf("%d",q[front++]);return 0;}
```

Day 84 - Linked List

```
#include <stdio.h>
#include <stdlib.h>
struct Node{int data;struct Node *next;};
int main(){struct Node *n=malloc(sizeof(struct Node));n->data=10;n->next=NULL;printf("%d",n->data);free(n);return 0;}
```

Day 85 - Linked List Traversal

```
#include <stdio.h>
struct Node{int data;struct Node *next;};
int main(){struct Node a={10,NULL},b={20,NULL};a.next=&b;printf("%d %d",a.data,a.next->data);return 0;}
```

Day 86 - Search Array

```
#include <stdio.h>
int main(){int a[]={1,2,3,4,5},key=4;for(int i=0;i<5;i++)if(a[i]==key)printf("Found at %d",i);return 0;}
```

Day 87 - Second Largest

```
#include <stdio.h>
int main(){int a[]={10,30,20,50,40},max=0,second=0;for(int i=0;i<5;i++){if(a[i]>max){second=max;max=a[i];}else if(a[i]>second){second=a[i];}}
```

Day 88 - Remove Duplicate

```
#include <stdio.h>
int main(){int a[]={1,2,2,3,3},n=5;for(int i=0;i<n;i++){int d=0;for(int j=0;j<i;j++)if(a[i]==a[j])d=1;if(!d)printf("%d",a[i]);}}
```

Day 89 - Matrix Multiplication

```
#include <stdio.h>
int main(){int a[2][2]={1,2},{3,4}},b[2][2]={5,6},{7,8}},c[2][2]={0};for(int i=0;i<2;i++)for(int j=0;j<2;j++)for(int k=0;k<2;k++)c[i][j]=a[i][k]*b[k][j];for(int i=0;i<2;i++)for(int j=0;j<2;j++)printf("%d\t",c[i][j]);}
```

Day 90 - String Reverse

```
#include <stdio.h>
#include <string.h>
int main(){char s[]="Hello";for(int i=(int)strlen(s)-1;i>=0;i--)printf("%c",s[i]);return 0;}
```

Day 91 - Character Count

```
#include <stdio.h>
int main(){char s[]="hello";int c=0;for(int i=0;s[i];i++)c++;printf("%d",c);return 0;}
```

Day 92 - Vowel Count

```
#include <stdio.h>
int main(){char s[]="education";int c=0;for(int i=0;s[i];i++)if(s[i]=='a' || s[i]=='e' || s[i]=='i' || s[i]=='o' || s[i]=='u')c++;
```

Day 93 - Swap Without Third Variable

```
#include <stdio.h>
int main(){int a=10,b=20;a=a+b;b=a-b;a=a-b;printf("%d %d",a,b);return 0;}
```

Day 94 - Decimal to Binary

```
#include <stdio.h>
int main(){int n=10,b[10],i=0;while(n){b[i++]=n%2;n/=2;}while(i-->0)printf("%d",b[i]);return 0;}
```

Day 95 - Binary to Decimal

```
#include <stdio.h>
int main(){int n=1010,d=0,p=1;while(n){d+=(n%10)*p;p*=2;n/=10;}printf("%d",d);return 0;}
```

Day 96 - Simple Calculator

```
#include <stdio.h>
int main(){char op;float a,b;scanf("%f %c %f",&a,&op,&b);if(op=='+')printf("%.2f",a+b);else if(op=='-')printf("%.2f",a-b);else if(op=='*')printf("%.2f",a*b);else if(op=='/')printf("%.2f",a/b);return 0;}
```

Day 97 - Student Grade

```
#include <stdio.h>
int main(){int marks=85;if(marks>=90)printf("A+");else if(marks>=80)printf("A");else if(marks>=70)printf("B");else if(marks>=60)printf("C");else if(marks>=50)printf("D");else printf("F");return 0;}
```

Day 98 - Mini Project

```
#include <stdio.h>
int main(){int a,b;printf("Enter two numbers: ");scanf("%d%d",&a,&b);printf("Sum = %d",a+b);return 0;}
```

Day 99 - C Revision

```
#include <stdio.h>
int main(){int n=5,sum=0;for(int i=1;i<=n;i++)sum+=i;printf("Sum = %d",sum);return 0;}
```

Day 100 - Final C Project

```
#include <stdio.h>
int main(){int a,b;printf("Enter two numbers: ");scanf("%d%d",&a,&b);printf("Addition = %d\n",a+b);printf("Subtraction = %d\n",a-b);printf("Multiplication = %d\n",a*b);printf("Division = %d\n",a/b);return 0;}
```